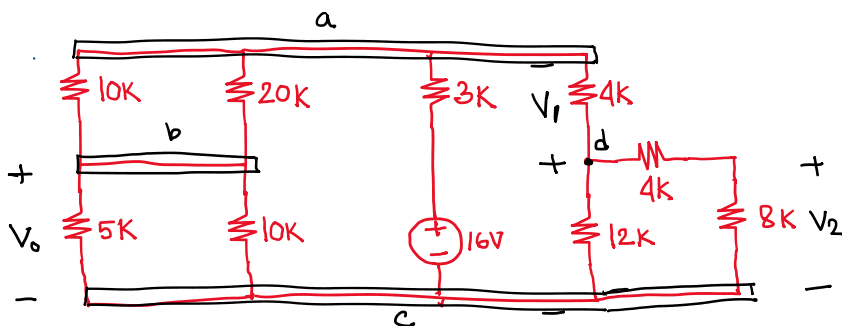
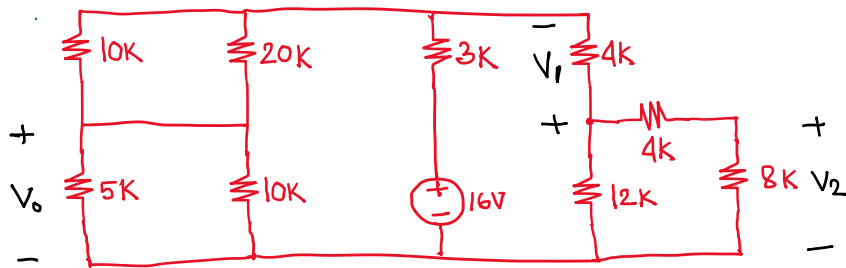


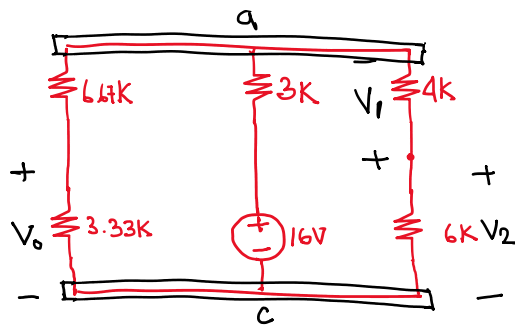


Between node a & b: $10K \parallel 20K$
 $= \frac{10 \times 20}{30}$
 $= 6.67K$

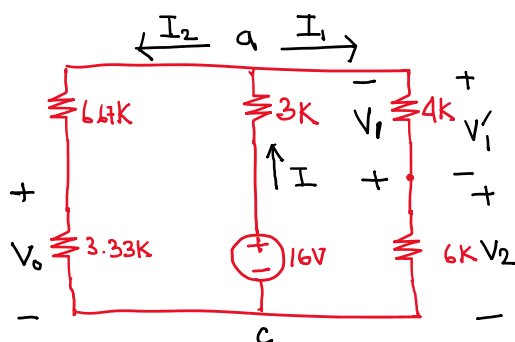
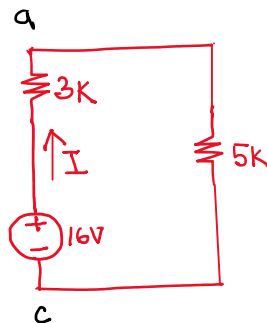
Between node c & b: $5K \parallel 10K$
 $= 3.33K$

Between node d & c: $12K \parallel (4K + 8K)$ ← Series
 $= 12K \parallel 12K$
 $= 6K$

Between node a & c: $(6.67K + 3.33K) \parallel (4K + 6K)$
 $= 10K \parallel 10K$
 $= 5K$



$$I = \frac{16}{8} = 2mA$$



Using Current Divider Rule:

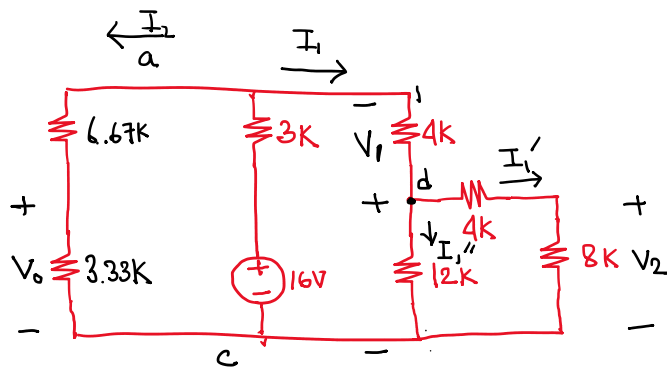
$$I_1 = \frac{10}{20} \times 2 = 1mA$$

$$I_2 = \frac{10}{20} \times 2 = 1mA$$

$$V_1' = 4K \times I_1 = 4K \times 1mA = 4V$$

$$V_1 = -V_1' = -4V$$

$$V_0 = I_2 \times 3.33K = 1mA \times 3.33K = 3.33V$$

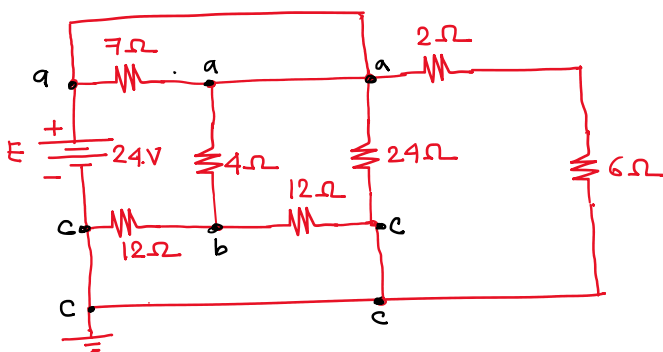
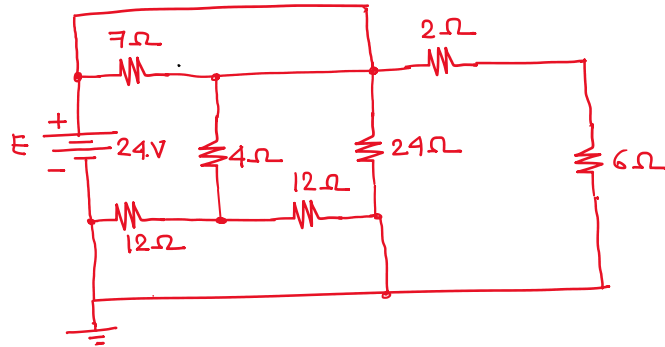
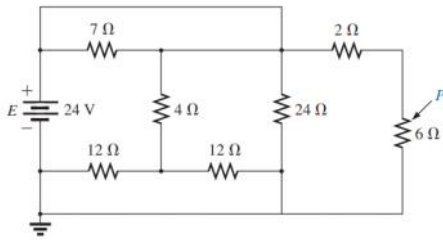


Applying current divider rule
at d

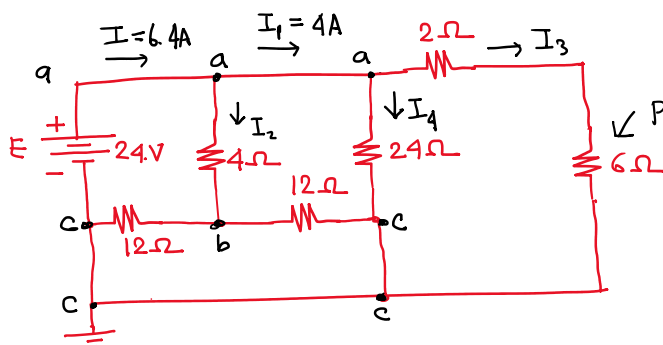
$$I_1' = \frac{12}{12+4+8} I_1 = \frac{12}{24} \times 1 \text{ mA} = 0.5 \text{ mA}$$

$$V_2 = 8 \times I_1' = 8 \times 0.5 = 4 \text{ V}$$

*36. Determine the power delivered to the 6 Ω load in Fig. 7.101.

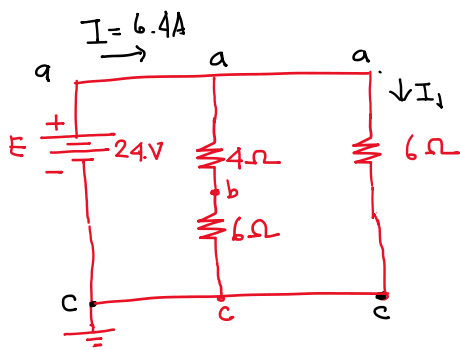


7 Ω Resistor shorted out

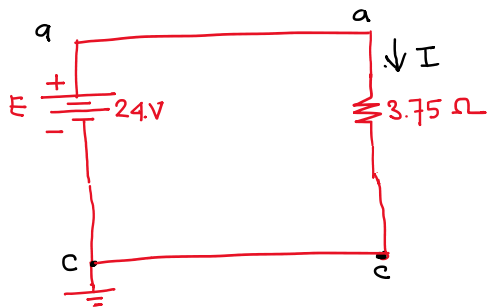


Between node b & c: $12 \Omega \parallel 12 \Omega = 6 \Omega$

Between node a & c: $24 \Omega \parallel (2 + 6) \Omega$
 $= 24 \Omega \parallel 8 \Omega$
 $= 6 \Omega$



Between node a & c: $(4+6)\Omega \parallel 6\Omega$
 $= 3.75\Omega$



$$I = \frac{E}{R_T} = \frac{24}{3.75} = 6.4 \text{ A}$$

$$I_1 = \frac{10}{16} \times 6.4 \text{ A} = 4 \text{ A}$$

$$I_3 = \frac{24}{32} \times 4 \text{ A} = 3 \text{ A}$$

Power absorbed by 6Ω Resistor
 $= I_3^2 \times 6 = 3^2 \times 6 = 9 \times 6 = 54 \text{ W}$